

**INSTITUTO
FEDERAL**
Santa Catarina

Introdução à Programação Web

Programação Frontend I

Análise e Desenvolvimento de Sistemas

Prof. Adriano Lima

adriano.lima@ifsc.edu.br



INSTITUTO FEDERAL

Santa Catarina

Câmpus São José

PENSAMENTO
COMPUTACIONAL E
ALGORITMOS

Análise e Desenvolvimento
de Sistemas

Programação Frontend



frontend



backend

Programação Frontend



frontend

- "o lado do cliente"



backend

- "o lado do servidor"

Programação Frontend



frontend

- "o lado do cliente"
- texto, cores, imagens



backend

- "o lado do servidor"
- transforma e armazena dados

Programação Frontend



frontend

- "o lado do cliente"
- texto, cores, imagens
- HTML, CSS, e JavaScript



backend

- "o lado do servidor"
- transforma e armazena dados
- escalabilidade

Programação Frontend



frontend

- "o lado do cliente"
- texto, cores, imagens
- HTML, CSS, e JavaScript
- usabilidade, acessibilidade, responsividade



backend

- "o lado do servidor"
- transforma e armazena dados
- escalabilidade
- segurança

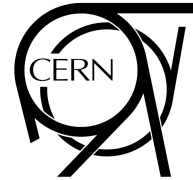
Histórico



Tim Berners-Lee



Robert Cailliau



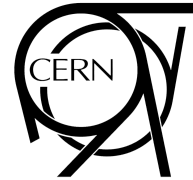
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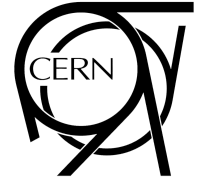


Tim Berners-Lee

1989 Proposta da arquitetura



Robert Cailliau



Histórico



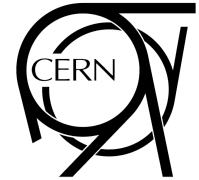
Tim Berners-Lee



Robert Cailliau

1989 Proposta da arquitetura

1990 HTTP, HTML
1º web browser (WorldWideWeb)
1º web server (CERN httpd)
1º web site (<http://info.cern.ch>)



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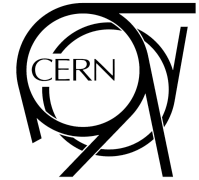


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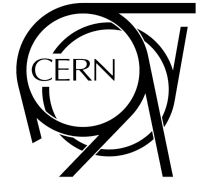
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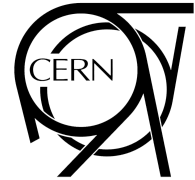


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- 1993** 1º navegador gráfico (NCSA Mosaic)
- 1994** World Wide Web Consortium (W3C)



Primeiro Website

```
window['_gaUserPrefs'] = { 100 : function() { return true; } }  
[touch-action='none']{ -ms-touch-action: none; touch-action: none; }[touch-  
action='pan-x']{ -ms-touch-action: pan-x; touch-action: pan-x; }[touch-  
action='pan-y']{ -ms-touch-action: pan-y; touch-action: pan-y; }[touch-  
action='scroll'],[touch-action='pan-x pan-y'],[touch-action='pan-y pan-x']{ -ms-  
touch-action: pan-x pan-y; touch-action: pan-x pan-y; }
```

The World Wide Web project

WORLD WIDE WEB

The WorldWideWeb (W3) is a wide-area hypermedia[1] information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an executive summary[2] of the project, Mailing lists[3] , Policy[4] , November's W3 news[5] , Frequently Asked Questions[6] .

What's out there?[7]Pointers to the world's online information, subjects[8] , W3 servers[9], etc.

Help[10] on the browser you are using

Software A list of W3 project components and their current
Products[11] state. (e.g. Line Model[12] ,X11 Viola[13] ,

<ref.number>, Back, <RETURN> for more, or Help: █



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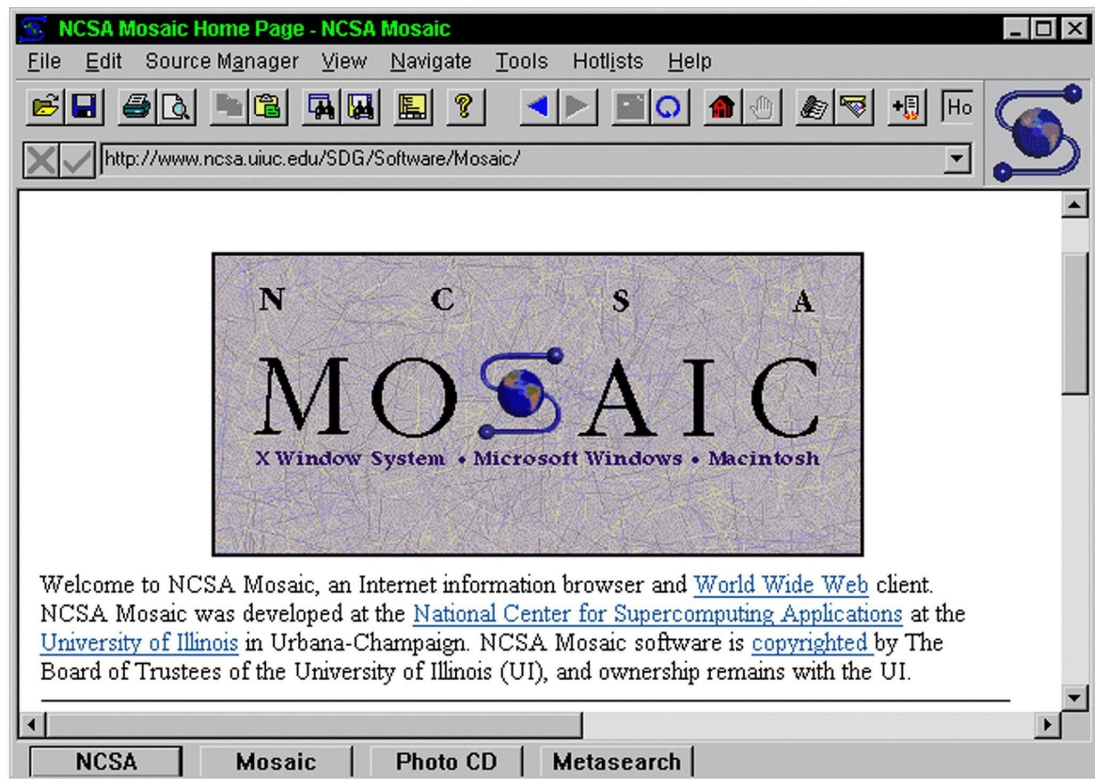
Software

Products[11]

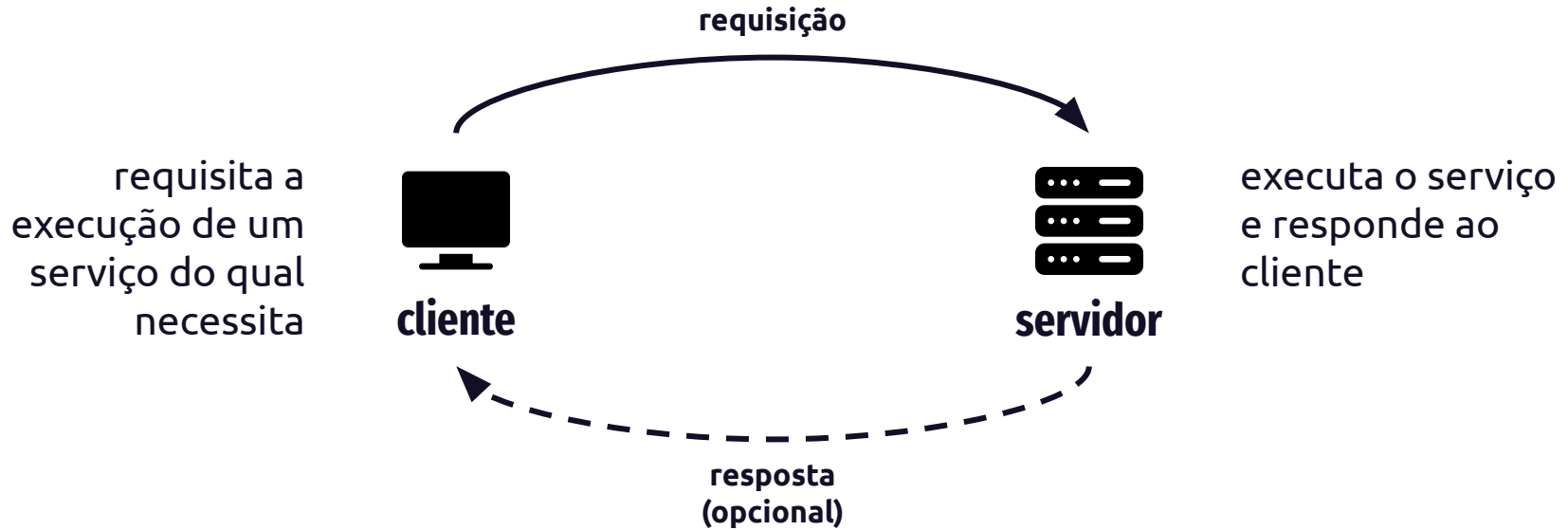
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NCSA Mosaic



Modelo Cliente-Servidor



Modelo Cliente-Servidor

modelo mais utilizado para interação entre computadores em rede



Modelo Cliente-Servidor

modelo mais utilizado para interação entre computadores em rede



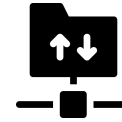
adotado por diversos serviços da internet



web



email



FTP



telnet

...

Modelo Cliente-Servidor

modelo mais utilizado para interação entre computadores em rede



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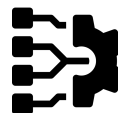
empregado na construção de diversos tipos de sistemas de informação



BD



CRM



ERP

...

Modelo Cliente-Servidor



cliente



servidor

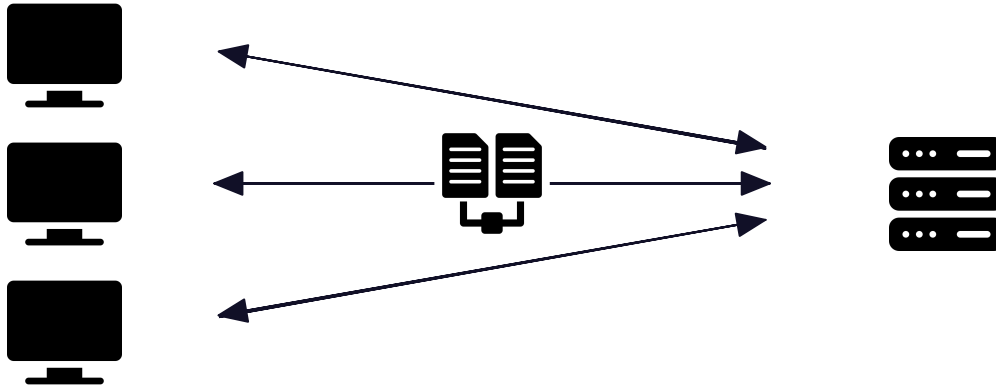
papéis

- | | |
|---|---|
| ■ solicitar a execução do serviço | ■ conhecer a lógica de execução do serviço |
| ■ fornecer parâmetros de execução | ■ ter acesso aos recursos necessários para execução |
| ■ exibir, armazenar ou utilizar o resultado enviado pelo servidor | |

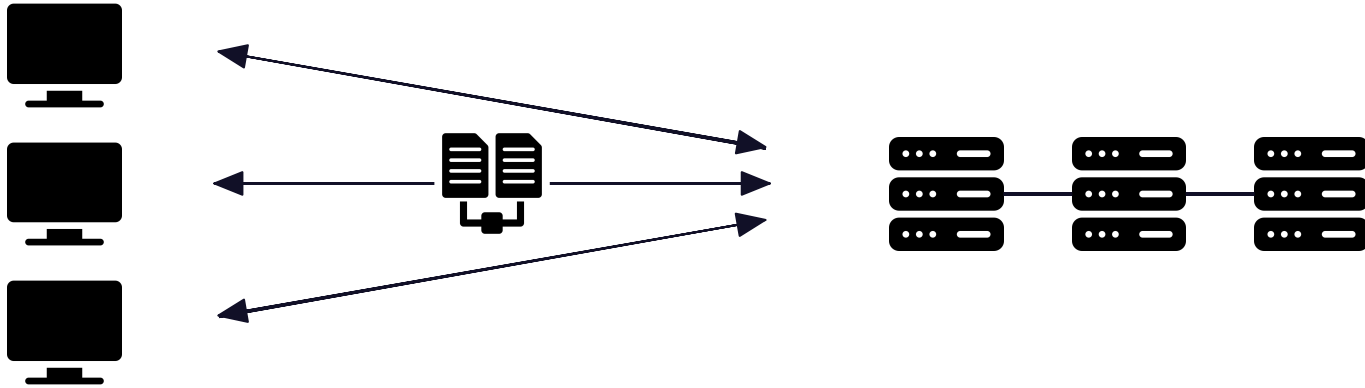
Modelo Cliente-Servidor



Modelo Cliente-Servidor



Modelo Cliente-Servidor



Arquitetura



modelo cliente-servidor

protocolo de comunicação http (hypertext transfer protocol)

outros protocolos aceitos ftp, nntp, mailto, ...

formato do conteúdo html (hypertext markup language)

plataforma dos navegadores pc, celular, tablet, tv, ...

s.o. dos servidores linux, windows, mac



Arquitetura



navegadores web

acessam servidores web e
solicitam conteúdos

exibem para o usuário
o conteúdo acessado

permitem que o usuário navegue
para outros conteúdos
da web acionando hyperlinks

servidores web

disponibilizam conteúdo
na web

conteúdo armazenado
estaticamente ou gerado
dinamicamente



Navegadores



chrome



safari

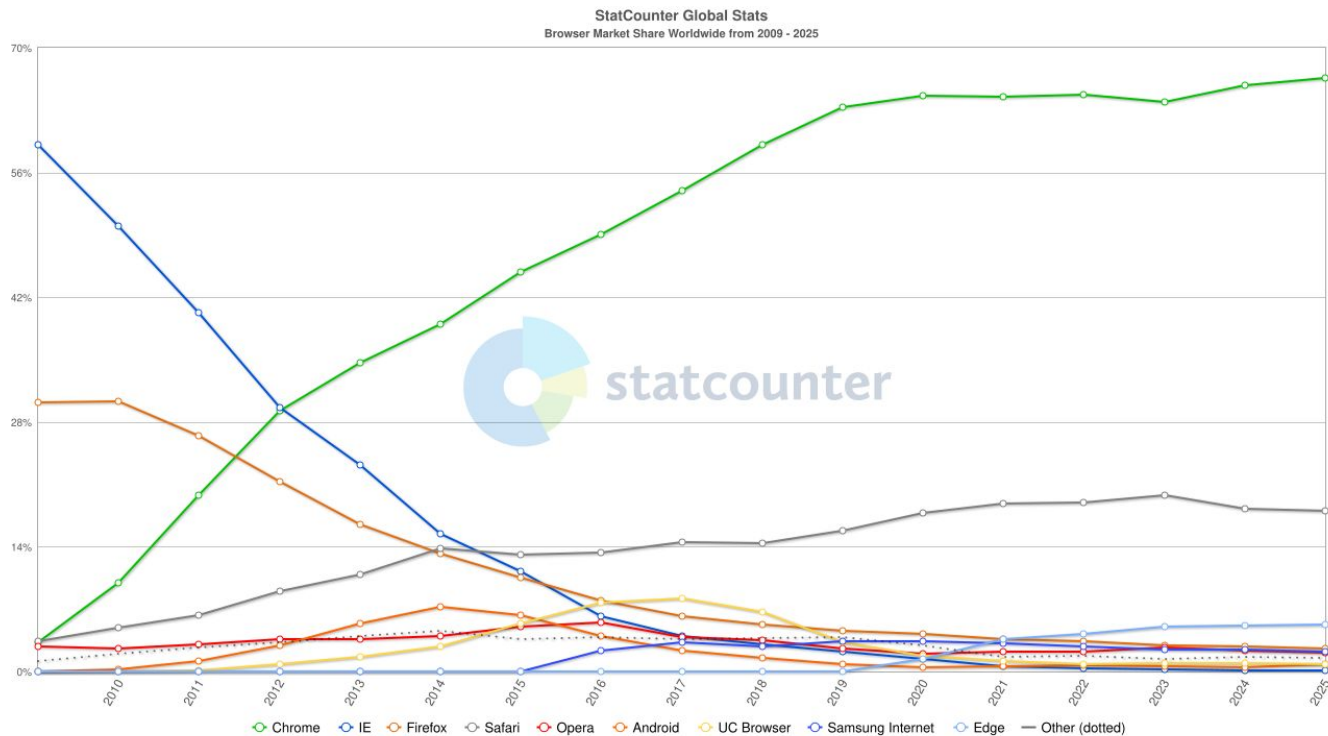


edge



firefox

Navegadores



Navegadores

Ctrl+U

The image shows a side-by-side comparison of a web browser window. On the left, the browser displays the homepage of the Instituto Federal de Santa Catarina (IFSC). The page features a header with the IFSC logo and the text 'INSTITUTO FEDERAL Santa Catarina'. Below the header, there is a large banner image with the text 'Vagas remanescentes para cursos técnicos Proeja' and 'Inscreva-se até 16 de fevereiro'. The browser's address bar shows 'https://www.ifsc.edu.br/'. On the right, the same browser window is shown with the source code of the page displayed. The source code is in HTML and includes a title 'Página Inicial - Portal do IFSC', a meta tag for the viewport, and links to Leaflet CSS and JS. The browser's address bar shows 'view-source:https://www.ifsc.edu.br/'.

File Edit View History Bookmarks Tools Help

Página Inicial - Portal do IFSC

https://www.ifsc.edu.br/

BRASIL

INSTITUTO FEDERAL Santa Catarina

Vagas remanescentes para cursos técnicos Proeja

Inscreva-se até 16 de fevereiro

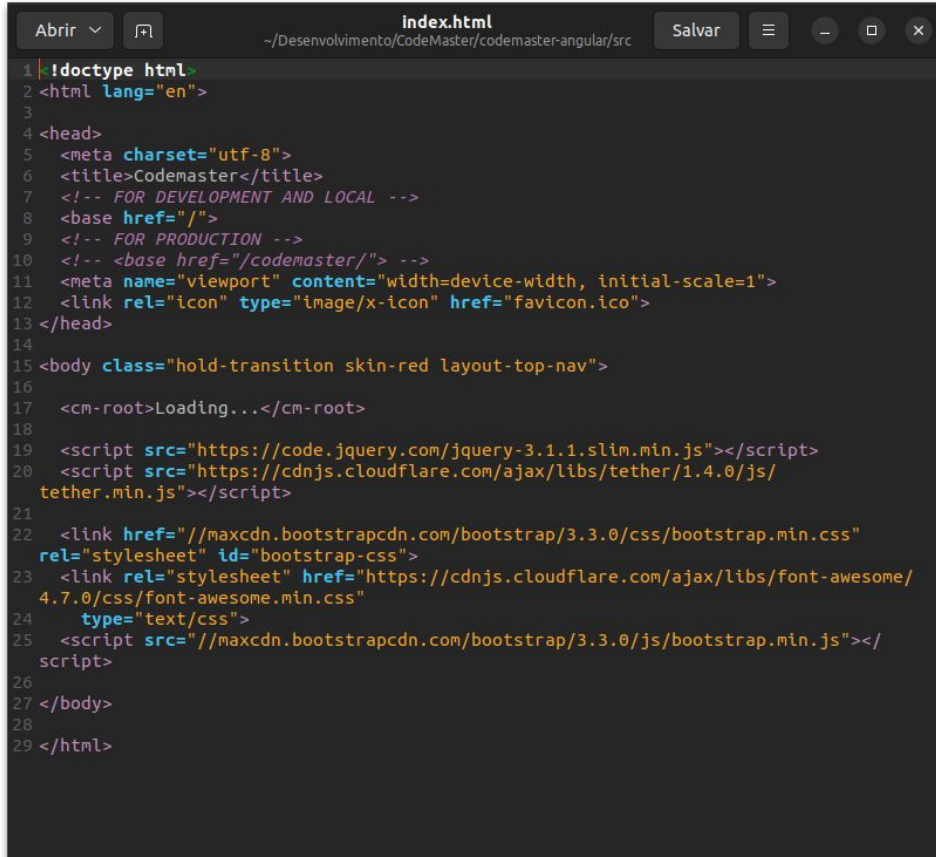
view-source:https://www.ifsc.edu.br/

```
84
85
86
87
88
89
90
91
92
93
94
95
96
97 <html class="ltr" dir="ltr" lang="pt-BR">
98
99 <head>
100   <title>Página Inicial - Portal do IFSC</title>
101
102   <meta content="initial-scale=1.0, width=device-width" name="viewport" />
103   <link
104     rel="stylesheet"
105     href="https://unpkg.com/leaflet@1.7.1/dist/leaflet.css"
106     integrity="sha512-xodZBNTCSn17Xt2atTPuE1HxjVMSvLVW9ocqUKLsCC5CXdbqCmblAsh0MAS6/keqq/sMZMZ19scR4PsZChSR7A=="
107     crossorigin="" />
108
109   <!-- Make sure you put this AFTER Leaflet's CSS -->
110   <script
111     src="https://unpkg.com/leaflet@1.7.1/dist/leaflet.js"
112     integrity="sha512-XQoYMqMTK8LvdxXYG3nZ448hOEQiglffqkJs1NOqV44cWnUrBc8PkA0cXy20w0vLaXaVUearIOBhiXZ5V3ynxwA=="
113     crossorigin=""
114   />
115
116   <!-- Global site tag (gtag.js) - Google Analytics -->
```

**Ctrl+Shift+I
ou F12**



Ferramentas de Desenvolvimento



```
1 |<!doctype html>
2 <html lang="en">
3
4 <head>
5   <meta charset="utf-8">
6   <title>Codemaster</title>
7   <!-- FOR DEVELOPMENT AND LOCAL -->
8   <base href="/">
9   <!-- FOR PRODUCTION -->
10  <!-- <base href="/codemaster/"> -->
11  <meta name="viewport" content="width=device-width, initial-scale=1">
12  <link rel="icon" type="image/x-icon" href="favicon.ico">
13 </head>
14
15 <body class="hold-transition skin-red layout-top-nav">
16
17   <cm-root>Loading...</cm-root>
18
19   <script src="https://code.jquery.com/jquery-3.1.1.slim.min.js"></script>
20   <script src="https://cdnjs.cloudflare.com/ajax/libs/tether/1.4.0/js/
21   tether.min.js"></script>
22
23   <link href="//maxcdn.bootstrapcdn.com/bootstrap/3.3.0/css/bootstrap.min.css"
24   rel="stylesheet" id="bootstrap-css">
25   <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/
26   4.7.0/css/font-awesome.min.css"
27   type="text/css">
28   <script src="//maxcdn.bootstrapcdn.com/bootstrap/3.3.0/js/bootstrap.min.js"></
29   script>
30 </body>
31 </html>
```

editor de texto



TextEdit

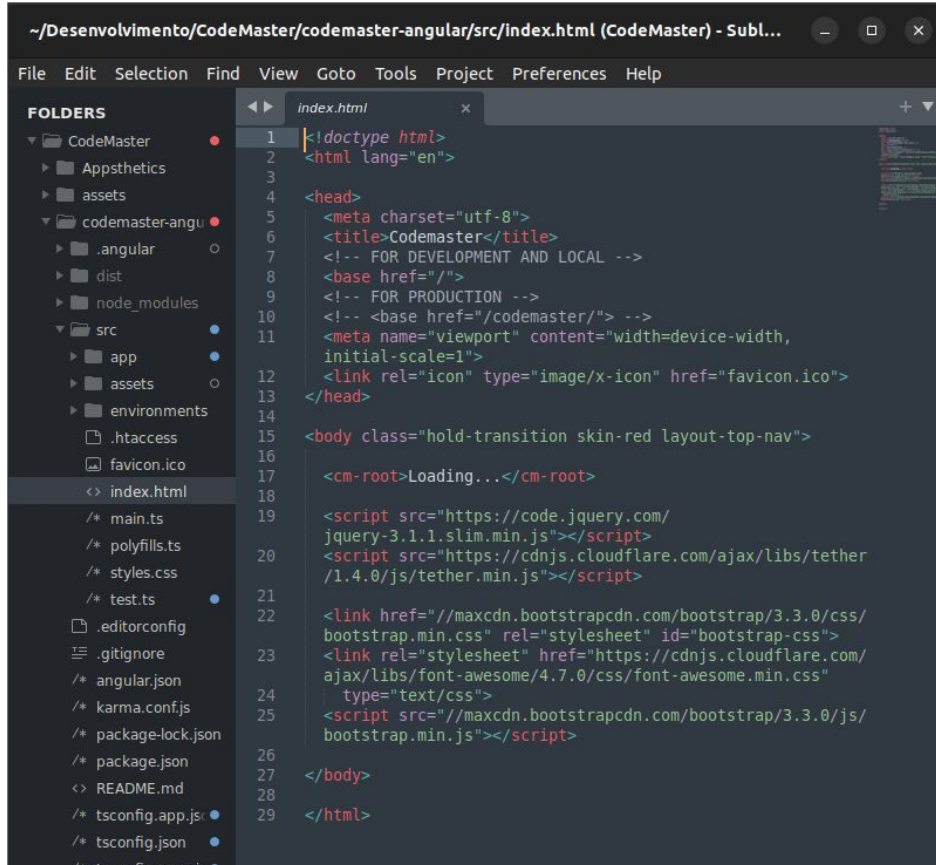


NotePad



Gedit

Ferramentas de Desenvolvimento



editor de código



Atom

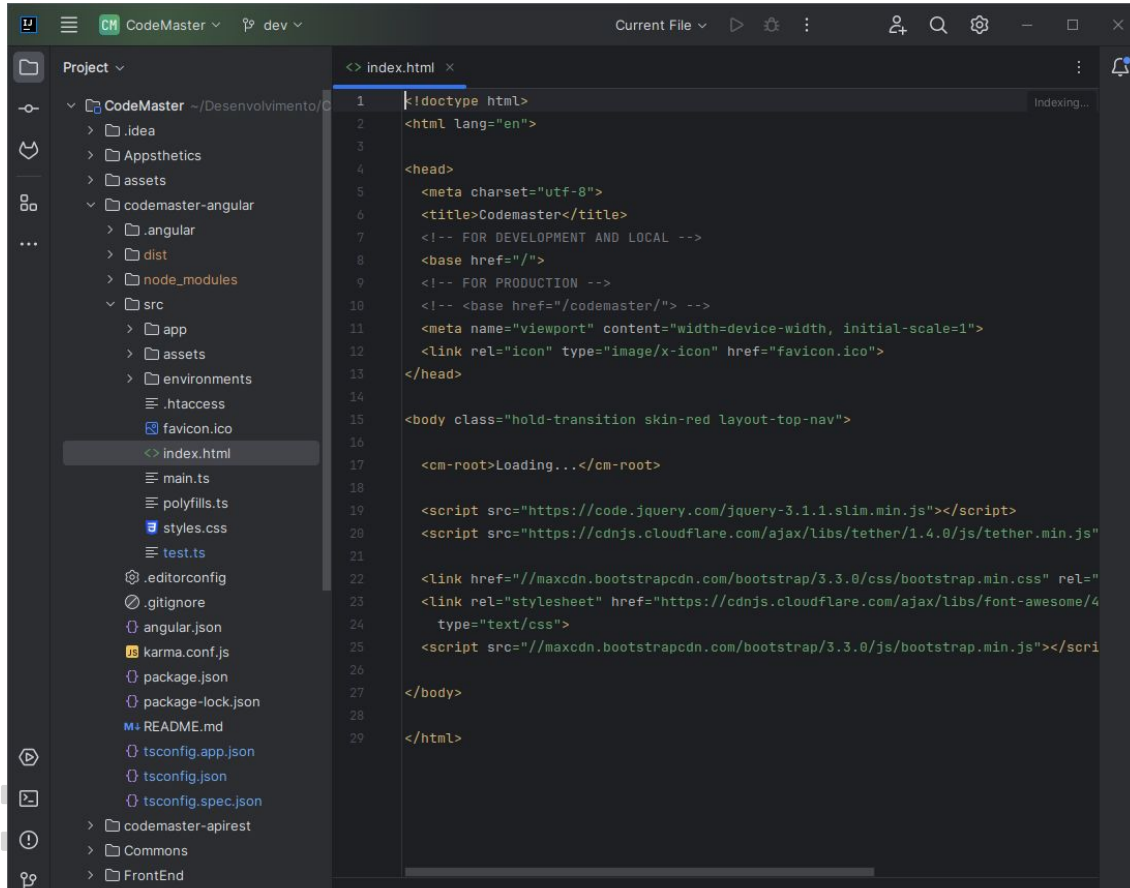


Sublime

TO
NAL E
DS

Análise e Desenvolvimento
de Sistemas

Ferramentas de Desenvolvimento



ambiente integrado de desenvolvimento

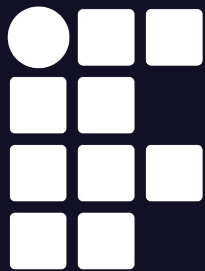


Visual Studio



IntelliJ IDEA

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